

**ADV: Trigonometry (Adv), T2 Trig Functions and Identities (Adv)
Trig Identities and Harder Equations (Y11)**

Teacher: Ms Najem

Exam Equivalent Time: 54 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

1. Trigonometry, 2ADV T2 2017 HSC 7 MC

Which expression is equivalent to $\tan \theta + \cot \theta$?

- (A) $\operatorname{cosec} \theta + \sec \theta$
- (B) $\sec \theta \operatorname{cosec} \theta$
- (C) 2
- (D) 1

2. Trigonometry, 2ADV T2 SM-Bank 44 MC

The implied domain of the function with rule $f(x) = 1 - \sec\left(x + \frac{\pi}{4}\right)$ is

- A. all real x
- B. $\left\{\frac{(4k-1)\pi}{4}\right\}$, (for k integer)
- C. $\left\{\frac{(4k+1)\pi}{4}\right\}$, (for k integer)
- D. $\left\{\frac{(2k-1)\pi}{4}\right\}$, (for k integer)

3. Trigonometry, 2ADV T2 2014 HSC 7 MC

How many solutions of the equation $(\sin x - 1)(\tan x + 2) = 0$ lie between 0 and 2π ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

4. Trigonometry, 2ADV T2 2016 HSC 8 MC

How many solutions does the equation $|\cos(2x)| = 1$ have for $0 \leq x \leq 2\pi$?

- (A) 1
- (B) 3
- (C) 4
- (D) 5

5. Trigonometry, 2ADV T2 SM-Bank 41

Prove that

$$(\sec x + \tan x)(\sec x - \tan x) = 1. \quad (2 \text{ marks})$$

6. Trigonometry, 2ADV T2 SM-Bank 2

Find all solutions of the equation $2\cos \theta = \sqrt{3}\cot \theta$, for $0 \leq \theta \leq 2\pi$ (3 marks)

7. Trigonometry, 2ADV T2 SM-Bank 32

Express $5\cot^2 x - 2\operatorname{cosec} x + 2$ in terms of $\operatorname{cosec} x$ and hence solve

$$5\cot^2 x - 2\operatorname{cosec} x + 2 = 0 \text{ for } 0 < x < 2\pi. \quad (3 \text{ marks})$$

8. Trigonometry, 2ADV T2 SM-Bank 34

Solve the equation $\sqrt{3}\sin x = \cos x$ for $-\pi \leq x \leq \pi$. (2 marks)

9. Trigonometry, 2ADV T2 SM-Bank 42

Prove that

$$\frac{1 - \sin^2 x \cos^2 x}{\cos^2 x} = \cot^2 x + \sin^2 x. \quad (2 \text{ marks})$$

10. Calculus, 2ADV C4 2010 HSC 5b

(i) Prove that

$$\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x}. \quad (1 \text{ mark})$$

(ii) Hence prove that

$$\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}. \quad (1 \text{ mark})$$

(iii) Hence, use the identity $\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax$ to find the exact value of

$$\int_0^{\frac{\pi}{4}} \frac{1}{1 - \sin x} \, dx. \quad (2 \text{ marks})$$

11. Trigonometry, 2ADV T2 2004 HSC 8a

(i) Show that $\cos \theta \tan \theta = \sin \theta$. *(1 mark)*

(ii) Hence solve $8 \sin \theta \cos \theta \tan \theta = \operatorname{cosec} \theta$ for $0 \leq \theta \leq 2\pi$. *(2 marks)*

12. Trigonometry, 2ADV T2 2004 HSC 9a

Consider the geometric series $1 - \tan^2 \theta + \tan^4 \theta - \dots$

(i) When the limiting sum exists, find its value in simplest form. *(2 marks)*

(ii) For what values of θ in the interval

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2} \text{ does the limiting sum of the series exist? } \quad (2 \text{ marks})$$

13. Trigonometry, 2ADV T2 2014 HSC 15a

Find all solutions of $2 \sin^2 x + \cos x - 2 = 0$, where $0 \leq x \leq 2\pi$. *(3 marks)*

14. Trigonometry, 2ADV T2 2019 HSC 13a

Solve $2 \sin x \cos x = \sin x$ for $0 \leq x \leq 2\pi$. *(3 marks)*

15. Trigonometry, 2ADV T2 SM-Bank 40

Let $(\tan \theta - 1)(\sin \theta - \sqrt{3} \cos \theta)(\sin \theta + \sqrt{3} \cos \theta) = 0$.

a. State all possible values of $\tan \theta$. *(1 mark)*

b. Hence, find all possible solutions for $(\tan \theta - 1)(\sin^2 \theta - 3 \cos^2 \theta) = 0$, where $0 \leq \theta \leq \pi$.
(2 marks)

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Worked Solutions

1. Trigonometry, 2ADV T2 2017 HSC 7 MC

$$\begin{aligned}\tan \theta + \cot \theta &= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \\ &= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta} \\ &= \frac{1}{\cos \theta \sin \theta} \\ &= \sec \theta \operatorname{cosec} \theta\end{aligned}$$

$\Rightarrow B$

2. Trigonometry, 2ADV T2 SM-Bank 44 MC

$$y = \sec(x) = \frac{1}{\cos(x)} \text{ has asymptotes when}$$

$$x = -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$\Rightarrow y = \sec\left(x + \frac{\pi}{4}\right) \text{ has asymptotes at when}$$

$$x = \frac{-3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}, \dots$$

$$\therefore \text{Domain: } \left\{ \frac{(4k+1)\pi}{4} \right\}, (\text{for } k \text{ integer})$$

$\Rightarrow C$

3. Trigonometry, 2ADV T2 2014 HSC 7 MC

When $(\sin x - 1)(\tan x + 2) = 0$

$(\sin x - 1) = 0$ or $\tan x + 2 = 0$

If $\sin x - 1 = 0$

$\sin x = 1$

$x = \frac{\pi}{2}, \quad 0 < x < 2\pi$

If $\tan x + 2 = 0$

$\tan x = -2$

$\Rightarrow$ Note that since $\tan \frac{\pi}{2}$ is undefined, there

are only 2 solutions when $\tan x = -2$

(which occurs in the 1st and 4th quadrants).

$\therefore$ 2 solutions

$\Rightarrow B$

♦♦♦ Mean mark 25%, making it the toughest MC question in the 2014 exam.

COMMENT: Note that the "2 solutions" answer relies on the sum of an infinity of zeros not equalling zero. This concept created unintended difficulty in this question.

4. Trigonometry, 2ADV T2 2016 HSC 8 MC

$|\cos(2x)| = 1$

$\cos(2x) = \pm 1$

When $\cos(2x) = 1$

$2x = 0, 2\pi, 4\pi, \dots$

$\therefore x = 0, \pi, 2\pi, \dots$

When $\cos(2x) = -1$

$2x = \pi, 3\pi, 5\pi, \dots$

$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

$\therefore x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ for $0 \leq x \leq 2\pi$

$\Rightarrow D$

♦♦♦ Mean mark 23%.

5. Trigonometry, 2ADV T2 SM-Bank 41

$LHS = (\sec x + \tan x)(\sec x - \tan x)$

$= \sec^2 x - \tan^2 x$

$= \frac{1}{\cos^2 x} - \frac{\sin^2 x}{\cos^2 x}$

$= \frac{1 - \sin^2 x}{\cos^2 x}$

$= \frac{\cos^2 x}{\cos^2 x}$

$= 1$

$= RHS$

6. Trigonometry, 2ADV T2 SM-Bank 2

$2 \cos \theta = \sqrt{3} \cot \theta$

$2 \cos \theta - \sqrt{3} \cot \theta = 0$

$2 \cos \theta - \sqrt{3} \frac{\cos \theta}{\sin \theta} = 0$

$\left(2 - \frac{\sqrt{3}}{\sin \theta}\right) \cos \theta = 0$

If $\cos \theta = 0$,

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

If $2 - \frac{\sqrt{3}}{\sin \theta} = 0 \Rightarrow \sin \theta = \frac{\sqrt{3}}{2}$

$\theta = \frac{\pi}{3}, \frac{2\pi}{3}$

7. Trigonometry, 2ADV T2 SM-Bank 32

$$\begin{aligned}\cot^2 x &= \frac{\cos^2 x}{\sin^2 x} \\ &= \frac{1 - \sin^2 x}{\sin^2 x} \\ &= \operatorname{cosec}^2 x - 1\end{aligned}$$

$$5 \cot^2 x - 2 \operatorname{cosec} x + 2 = 0$$

$$5(\operatorname{cosec}^2 x - 1) - 2 \operatorname{cosec} x + 2 = 0$$

$$5 \operatorname{cosec}^2 x - 2 \operatorname{cosec} x - 3 = 0$$

$$(5 \operatorname{cosec} x + 3)(\operatorname{cosec} x - 1) = 0$$

$$\operatorname{cosec} x = -\frac{3}{5} \quad \operatorname{cosec} x = 1$$

$$\sin x = -\frac{5}{3} \quad \sin x = 1$$

$$(\text{no solution}) \quad x = \frac{\pi}{2}$$

$$\therefore x = \frac{\pi}{2}$$

8. Trigonometry, 2ADV T2 SM-Bank 34

Divide both sides by $\cos x$:

$$\sqrt{3} \sin x = \cos x$$

$$\sqrt{3} \tan x = 1$$

$$\tan x = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \text{Base angle} = \frac{\pi}{6}$$

$$\therefore x = \frac{\pi}{6} \text{ or } -\frac{5\pi}{6}, \quad (-\pi \leq x \leq \pi)$$

MARKER'S COMMENT: Many students who found the base angle correctly could not solve within the restrictions.

9. Trigonometry, 2ADV T2 SM-Bank 42

$$\begin{aligned}\text{RHS} &= \frac{\cos^2 x}{\sin^2 x} + \sin^2 x \\ &= \frac{\cos^2 x + \sin^4 x}{\sin^2 x} \\ &= \frac{\cos^2 x + \sin^2 x(1 - \cos^2 x)}{\sin^2 x} \\ &= \frac{\cos^2 x + \sin^2 x - \sin^2 x \cos^2 x}{\sin^2 x} \\ &= \frac{1 - \sin^2 x \cos^2 x}{\sin^2 x} \\ &= \text{LHS}\end{aligned}$$

10. Calculus, 2ADV C4 2010 HSC 5b

(i) Need to prove

$$\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x}$$

$$\begin{aligned} \text{LHS} &= \sec^2 x + \sec x \tan x \\ &= \frac{1}{\cos^2 x} + \frac{1}{\cos x} \times \frac{\sin x}{\cos x} \\ &= \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} \\ &= \frac{1 + \sin x}{\cos^2 x} \\ &= \text{RHS} \quad \dots \text{ as required} \end{aligned}$$

(ii) Need to prove

$$\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}$$

$$\text{i.e. } \frac{1 + \sin x}{\cos^2 x} = \frac{1}{1 - \sin x} \quad (\text{part (i)})$$

$$\begin{aligned} \text{LHS} &= \frac{1 + \sin x}{\cos^2 x} \\ &= \frac{1 + \sin x}{1 - \sin^2 x} \\ &= \frac{1 + \sin x}{(1 - \sin x)(1 + \sin x)} \\ &= \frac{1}{1 - \sin x} \quad \dots \text{ as required} \end{aligned}$$

$$(iii) \int_0^{\frac{\pi}{4}} \frac{1}{1 - \sin x} dx$$

$$= \int_0^{\frac{\pi}{4}} (\sec^2 x + \sec x \tan x) dx$$

$$= [\tan x + \sec x]_0^{\frac{\pi}{4}}$$

$$= \left[\left(\tan\left(\frac{\pi}{4}\right) + \sec\left(\frac{\pi}{4}\right) \right) - (\tan 0 + \sec 0) \right]$$

$$= \left[\left(1 + \frac{1}{\cos(\frac{\pi}{4})} \right) - \left(0 + \frac{1}{\cos 0} \right) \right]$$

♦♦ Mean mark 31%.

♦ Mean mark 37%.

$$= 1 + \sqrt{2} - 1$$

$$= \sqrt{2}$$

11. Trigonometry, 2ADV T2 2004 HSC 8a

(i) Prove $\cos \theta \tan \theta = \sin \theta$

$$\begin{aligned} \text{LHS} &= \cos \theta \tan \theta \\ &= \cos \theta \left(\frac{\sin \theta}{\cos \theta} \right) \\ &= \sin \theta \\ &= \text{RHS} \end{aligned}$$

(ii) $8 \sin \theta \cos \theta \tan \theta = \operatorname{cosec} \theta$

$$\therefore 8 \sin \theta (\sin \theta) = \operatorname{cosec} \theta, \quad (\text{part (i)})$$

$$8 \sin^2 \theta = \frac{1}{\sin \theta}$$

$$8 \sin^3 \theta = 1$$

$$\sin^3 \theta = \frac{1}{8}$$

$$\sin \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}. \quad (\text{for } 0 \leq \theta \leq 2\pi)$$

12. Trigonometry, 2ADV T2 2004 HSC 9a

(i) $1 - \tan^2 \theta + \tan^4 \theta - \dots$

$$\Rightarrow \text{GP where } a = 1, r = \frac{T_2}{T_1} = -\tan^2 \theta$$

$$\begin{aligned} \therefore S_{\infty} &= \frac{1}{1 - (-\tan^2 \theta)} \\ &= \frac{1}{1 + \tan^2 \theta} \\ &= \frac{1}{\sec^2 \theta} \\ &= \cos^2 \theta \end{aligned}$$

(ii) Find θ such that $|r| < 1$

$$|-\tan^2 \theta| < 1$$

$$\tan^2 \theta < 1$$

$$-1 < \tan \theta < 1$$

$$\therefore -\frac{\pi}{4} < \theta < \frac{\pi}{4}$$

13. Trigonometry, 2ADV T2 2014 HSC 15a

$$2 \sin^2 x + \cos x - 2 = 0$$

$$2(1 - \cos^2 x) + \cos x - 2 = 0$$

$$2 - 2 \cos^2 x + \cos x - 2 = 0$$

$$-2 \cos^2 x + \cos x = 0$$

$$\cos x(-2 \cos x + 1) = 0$$

$$\therefore -2 \cos x + 1 = 0 \text{ or } \cos x = 0$$

$$2 \cos x = 1 \quad x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\cos x = \frac{1}{2}$$

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Since $\cos$ is positive in 1st/4th quadrants,

$$\begin{aligned} x &= \frac{\pi}{3}, 2\pi - \frac{\pi}{3} \\ &= \frac{\pi}{3}, \frac{5\pi}{3} \end{aligned}$$

$$\therefore x = \frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{3} \text{ for } 0 \leq x \leq 2\pi$$

♦ Mean mark 42%

14. Trigonometry, 2ADV T2 2019 HSC 13a

$$2 \sin x \cos x - \sin x = 0$$

$$\sin x(2 \cos x - 1) = 0$$

$$\sin x = 0$$

$$\Rightarrow x = 0, \pi, 2\pi$$

$$\cos x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

♦ Mean mark 49%

15. Trigonometry, 2ADV T2 SM-Bank 40

a. $(\tan \theta - 1)(\sin \theta - \sqrt{3} \cos \theta)(\sin \theta + \sqrt{3} \cos \theta) = 0$

$$\Rightarrow \tan \theta = 1$$

$$\Rightarrow \sin \theta - \sqrt{3} \cos \theta = 0$$

$$\sin \theta = \sqrt{3} \cos \theta$$

$$\tan \theta = \sqrt{3}$$

$$\Rightarrow \sin \theta + \sqrt{3} \cos \theta = 0$$

$$\sin \theta = -\sqrt{3} \cos \theta$$

$$\tan \theta = -\sqrt{3}$$

$$\therefore \tan \theta = 1 \text{ or } \tan \theta = \pm \sqrt{3}$$

b. $(\tan \theta - 1)(\sin^2 \theta - 3 \cos^2 \theta) = 0$

Using part a:

$$(\tan \theta - 1)(\sin \theta - \sqrt{3} \cos \theta)(\sin \theta + \sqrt{3} \cos \theta) = 0$$

$$\Rightarrow \tan \theta = 1 \quad \text{or} \quad \tan \theta = \pm \sqrt{3}$$

$$\theta = \frac{\pi}{4} \quad \theta = \frac{\pi}{3}, \frac{2\pi}{3}$$

$$\therefore \theta = \frac{\pi}{4}, \frac{\pi}{3} \text{ or } \frac{2\pi}{3} \quad (0 \leq \theta \leq \pi)$$

♦ Mean mark 42%.

♦ Mean mark 42%.